

OPEN SOURCE SINGLE BOARD COMPUTER

A complete computer on a single circuit board with processor, memory and input/output is termed as single board computer (SBC). Below are some explorable websites related to Open Source SBC.

beagleboard.org

BeagleBoard is a low-power, low-cost, fan-less single-board computer. It is based on Texas Instruments' processors featuring the ARM Cortex-A series core. BeagleBoard was designed with Open Source development in mind, as a way of demonstrating the TI's OMAP3530 system-on-chip.

The website is an all-volunteer-activity started by passionate individuals including TI employees interested in creating powerful, Open and embedded devices. The site has a discussion board along with links that provide information regarding where you can buy the board. It also has references to around 200 different projects related to the board.



URL: <http://beagleboard.org/>



hawkboard.org

Hawkboard is a single-board computer based on TI's multicore OMAP processors. It is made around 300MHz OMAP L138 that is based on a different ARM/TI combination—ARM 9 and floating point DSP C674x application processor. The website is an Open community portal for AM1808/OMAPL138 platform. It provides details on where to buy the board. It also has a complete user-guide, which can be downloaded in the PDF format. There is a link on the home page to the IRC channel of Hawkboard discussion forum.

URL: <http://www.hawkboard.org>

pandaboard.org

PandaBoard is a low-cost, low-power, single-board computer based on the TI's OMAP4430 processor. The website is an online resource and collaboration portal for the growing active community supporting PandaBoard. It offers useful resources for mobile application developers designing on Linux-based distributions. It also provides developers the opportunity to engage with community experts and source technical material. The site is managed by volunteers, who are committed to Open Source development.



URL: <http://www.pandaboard.org/>



gumstix.org

In 2003, Gumstix pioneered the concept of an extremely small computer-on-module with a full implementation of Linux. They provide some of the most innovative design and production solutions to design and build small devices. The site is a complete guide for using a Gumstix computer-on-module effectively and efficiently. The tutorials on the site explain how to boot and connect Gumstix computers to expansion cards. It also connects you to the Gumstix mailing list on sourceforge.net

URL: <http://www.gumstix.org/>

elinux.org

elinux.org is a wiki on embedded Linux. Its main purpose is to preserve and present information about the development and use of Linux in embedded systems. It has tutorials and information on SBC like Hawkboard and BeagleBoard. The main portals of the site—Boot Time, Memory Management, Security, Events, Multimedia, Hardware Hacking, etc.—take you to lists of resources or information you can use to tackle problems in the particular area referred to.



URL: http://elinux.org/Main_Page